SOLITUDE 16 MW SOLAR PV

Document Prepared by Earthood Services Private Limited

Project Title	SOLITUDE 16 MW SOLAR PV
Report Title	Joint Validation and Verification report for "SOLITUDE 16 MW SOLAR PV"
Version	01
Report ID	VCS.VAL.20.24
Verification Period	19/12/2018 to 30/04/2020
Client	Voltas Yellow Ltd.
Pages	45
Date of Issue	20/10/2020
Prepared By	Earthood Services Private Limited
Contact	Regd. Office: 409-410A, Tower B4, Spaze I-Tech Park, Sector 49, Sohna Road, Gurgaon-122018, INDIA Tel: +91 124 4204599 Fax: +91 124 4204599 Website: www.earthood.in Email:info@earthood.in
Approved By	Dr. Kaviraj Singh

Work Carried Out By	Team Leader: Deepika Mahala Technical Expert: Deepika Mahala Verifier: Rahi Sahni Technical Reviewer: Shreya Garg Expert to Technical Reviewer: Shreya Garg

Summary:

The project activity involves the generation of electricity through a greenfield solar photovoltaic plant with a capacity of 16.328 MW_{DC} in Solitude near the village of Triolet in the north of Mauritius, in the district of Pamplemousses covering an overall area of 21 hectares. The solar plant consists of 59,376 photovoltaic modules that supply the generated electricity to the Central Electricity Board (CEB) grid.

The project aims to reduce the dependency on fossil fuel powered power plants for electricity by substituting carbon intensive energy with solar energy. In the baseline scenario, electricity delivered to the grid by the project activity would have otherwise been generated by operation of grid-connected power plants and by addition of new generation sources. The baseline scenario is the same as the scenario existing prior to the implementation of the project activity.

The plant is estimated to result in emission reductions worth 28,984 tons CO_2eq per year and 202,889 tons CO_2eq during the entire crediting period of 7 years.

During the first verification conducted for this project alongside validation, actual emission reductions from the monitoring period 19/12/2018 to 30/04/2020 were calculated as 29,681 tCO_2.

Scope of validation and verification

Voltas Yellow Ltd. contracted ESPL to conduct the joint validation and verification of the project. The project is covered under sectoral scope 1- Energy Industries (renewable sources) based on the methodology ACM0002: Large-scale Consolidated Methodology: Grid-connected electricity generation from renewable sources, Version 20.0.

A total of 01 CL, 02 CARs and 01 FAR have been raised during the joint validation and verification process of the project activity and successfully closed.

Conclusion

ESPL has performed the validation and verification of the VCS project activity "Solitude 16 MW solar PV".

The VVB has confirmed that:

- the PA is in accordance with all relevant host country criteria (Mauritius) and VCS rules and requirements;
- the PA is in accordance with all conditions of the latest version of applied methodology ACM0002, Version 20.0: “Grid-connected renewable electricity generation from renewable sources”;
- the local stakeholders’ consultation has been performed in accordance with host country and VCS requirements;
- the environmental assessment is appropriate and sufficient;
- the monitoring plan is transparent and adequate;
- all information has been consistently applied in the VCS-PD;

The implementation of the project has been done as per description in the VCS-PD.

CONTENTS

1 INTRODUCTION

1.1 Objective

Voltas Yellow Ltd. has contracted ESPL to conduct the joint validation and verification of project activity “Solitude 16 MW solar PV” according to the requirements of the Verified Carbon Standard Version 4.0.

1.2 Scope and Criteria

The scope of the validation and verification is to establish/verify that:

- the latest available VCS- Joint PD and MR was used and correctly filled up;
- the project activity is in accordance with all relevant host country criteria (Mauritius);
- the project activity is in accordance with all relevant VCS rules and requirements;
- the project activity is in accordance with conditions of the applied methodology ACM0002: Large Consolidated methodology --- Version 20.0/4/.

The validation and verification of the of the project activity is based on the VCS- Joint PD and MR, estimated GHG emission reduction calculations and actual emission reductions calculations.

1.3 Level of Assurance

☐ Limited level of assurance

☒ Reasonable level of assurance

A draft joint validation and verification report that is prepared by assessment team is reviewed by an independent technical review team (one or more members) to confirm if the internal procedures established and implemented by Earthood are duly complied with and such opinion/conclusion is reached in an objective manner that complies with the applicable VCS requirements as appropriate. The technical review team is collectively required to possess the technical expertise of all the technical area/sectoral scope the project activity relates to. All team members of technical review team are independent of the validation and verification team. The report approved by Technical Manager is endorsed by Managing Director, who is overall responsible to ensure quality, before final release. The further details of applicable procedures and responsibilities about Earthood Quality Management System (QMS) are available on its website (www.earthood.in).

Earthood’s validation approach is based on the understanding of the risks associated with reporting of GHG emission data and the controls in place to mitigate these. Earthood planned and performed the validation by obtaining evidence and other information and explanations that Earthood considers necessary to give reasonable level of assurance that reported estimated GHG emission reductions are fairly stated.

In our opinion the estimated and actual GHG emissions reductions were calculated correctly on the basis of the approved baseline and monitoring methodology ACM0002, Version 20.0/4/, and the VCS standard, Version 4.0/7/.

The validation and verification body has achieved Reasonable level of assurance without onsite visit by assessing the project through alternative means. Please refer section 2.4 of this report for detailed justification.

1.4 Summary Description of the Project

'Solitude 16 MW solar PV' generates renewable solar electricity and supplies power to the Central Electricity Board (CEB) grid. The project activity is a 16.328 MWDC solar power plant with 59,376 solar panel modules installed on a 21 hectares site in Solitude near the village of Triolet in the north of Mauritius, in the district of Pamplemousses. The power generated by the grid will be replacing an equivalent amount of electricity from the grid system of Mauritius which is majorly dependent on fossil-fuel based grid imports for its electricity.

The project is based on sectoral scope 1: Energy Industries with ACM0002: Large-scale Consolidated methodology: Grid-connected electricity generation from renewable sources, Version 20.0/4/.

As per the estimations made at the time of installation, the project will achieve a total of 29,233 MWh per year during the overall crediting period of 7 years from 19/12/2018 to 18/12/2025, resulting in 28,984 tons CO2eq of emission reductions per year on an average. The joint verification report also covers the first monitoring period from 19/12/2018 to 30/04/2020 resulting in a reduction of total of 29,681 tCO2e GHG emissions.

Year	Baseline emissions or removals (tCO_2e)	Project emissions or removals (tCO_2e)	Leakage emissions (tCO_2e)	Net GHG emission reductions or removals (tCO_2e)
2018	878	-	-	878
2019	21,070	-	-	21,070
2020	7,733	-	-	7,733
Total	29,681	-	-	29,681

2 VALIDATION AND VERIFICATION PROCESS

2.1 Method and Criteria

The joint Validation and Verification, from Contract Review to Verification Report & Opinion, was conducted using Earthood internal procedures. The Project was verified against the latest requirements (Version 4.0) /7/ and guidance set out in VCS and CDM Standards as applicable.

The validation and verification of project activity process is conducted as per internal CDM Quality Manual and in accordance with criteria laid down by VCS. It includes the following steps:

- contract with PP for the scope and appointment of validation, verification team and technical review team;
- completeness check of VCS-joint PD and MR;
- desk review of joint PD & MR and estimated as well as actual GHG emission reduction calculations;
- reporting and closure of findings (CARs/CLs/FARs) and preparation of draft joint validation and verification report;
- independent technical review of the draft report and final/revised documentation (e.g., VCS-Joint PD and MR, corresponding estimated ER calculations sheet and evidences);
- issuance of the final validation and verification report to contracted PP (or authorized representatives).

No sampling was required to be used since all data was directly verified through evidences and supporting documents provided by PP.

2.2 Document Review

The joint validation and verification is performed primarily as a document review of the documents submitted at various stages of assessments. The review is performed by assessment team using dedicated protocols. The assessment team cross checks the information provided in the documents (Joint Project Description & Monitoring Report/1/) and information from sources other than those used, if available, and also conducts independent background investigations. Earthood conducted a desk review as under;

a) A review of the data and information presented to verify their completeness;

b) A review of the monitoring plan, the monitoring methodology including applicable tool(s) and, where applicable, the applied standardized baseline, paying particular attention to the frequency of measurements, the quality of metering equipment including calibration requirements, and the quality assurance and quality control procedures;

c) An evaluation of data management and the quality assurance and quality control system in the context of their influence on the generation and reporting of emission reductions.

2.3 Interviews

The site visit was not conducted for this validation and verification assessment due to the outbreak of global pandemic (caused by COVID-19) and increased risk of exposure and contraction due to travel. It was not possible for the VVB team to conduct personal interviews with the project management team on-site. Also, the international travel is banned in the country of validation and verification team's residence/32/ which restricted the team from conducting the site visit physically.

However, alternate means were selected to discuss project implementation with the project management team.

All the relevant documents related to the Project Activity were collected via email beforehand and the information associated with the verification of Project Activity implementation were collected through a remote audit conducted on 18/09/2020 via Skype. The O&M personnel at the plant site were interviewed via video call to confirm information regarding the installed technology, data recording and storage procedures, plant maintenance and staff competency. The name plate details covering the provider and the serial numbers of all the installed technology were reviewed by the assessment team during the video call.

The details of the telephonic interview and video inspection of the site are as follows -

No.	Interviewee			Date	Topics
	Last name	First name	Affiliation		
1.	François	Vincent	Indian Ocean & Southern Africa- Solar Engineering and O&M Manager, GreenYellow	18/09/2020	Installed capacity and technological details of the plant, data recording and storage, monitoring management, and staff competency.
2.	Dunod	Alexandre	Director of Operations, AERA Group	18/09/2020	Monitoring management and data collection system

3.	Josian	Arlanda	GreenYellow site manager	18/09/2020	Site operations and monitoring management

The assessment team was able to procure all the relevant information related to the Project Activity through the remote survey. The project activity is also registered in CDM. So, the validation report was also referred to corroborate certain assumptions. The representative present at the plant site guided the assessment team through installed solar modules, inverters, transformers, metering system and through the data processing procedures in the plant office. Through the cross-checking of the information obtained during the remote survey against the evidences provided the assessment team is able to give a reasonable level of assurance that the all the information related to the Project Activity is transparent and justified.

2.4 Site Inspections

The validation and verification manual version 3.2/8/states that “physically observed data is considered most reliable, and that VVBs are required to provide a reasonable level of assurance, site visits must be included in validation and verification plans”.

The on-site inspection of the Project Activity could not be conducted due to outbreak of global pandemic Covid-19 and an increased risk of exposure and contraction due to travel. The country of residence for the VVB has the second highest number of cases in the world which is increasing with each passing day/34/. Moreover, International travel is also suspended in India (country of resince for VVB) till 31 October 2020/32/.

VERRA had launched travel guidance on 18/03/2020 (https://verra.org/covid-19-travel-guidance/) /30/, according to which Verra suggests to postponement of the site visit and informs that it will handle exemptions to our site visit requirements and validation/verification deadlines on a case-by-case basis. In absence of any new guideline from VERRA this guidance released on 18/03/2020 is still valid.

The site-visit for this verification could not be postponed as this would lead to delayed issuance of the CERs. The PP relies upon the CER revenue generated from the project as the working capital of the project.

Thus, the VVB performed the Joint Validation and Verification of the Project Activity through the adoption of alternative means to reach a reasonable level of assurance. The means of validation and verification are discussed in detail in relevant sections of this report.

The project is already CDM registered and all the documents used like Land lease document/10/, ESPA/16/ Commercial Operation Date certificate/14/, Design Compliance Report/27/ are approved or signed by Government body(as the electricity is being sold to Grid – central electricity board). The calibration was also maintained by Central Electricity

Board meter upon installation and at least once every year thereafter. In addition to this a video call was arranged with the team on ground through which all the nameplates were checked and matched with the information presented in the joint validation and verification report.

Thus, reasonable level of assurance was achieved in project implementation assessment.

Additionally, in line with Section 4.1.8 (1) of the VCS Standard, FAR 01 has been raised to ensure that all the relevant information related to the Project Activity shall be confirmed by the subsequent VVB during the second verification.

Therefore, it is concluded that the alternative means of validation and verification applied are sufficient to perform the Joint Validation and Verification of the Project Activity

2.5 Resolution of Findings

The findings may be of the following types: CAR- Corrective Action Request, CL- Clarification Request and FAR- Forward Action Request.

During the present joint validation and verification, 01 CL and 02 CARs were raised and successfully closed.

The list of findings and their resolution are presented in the Appendix IV of this verification report. The section also includes the response, if provided, by the project participants and an assessment by the verification team if it was closed our or otherwise.

2.5.1 Forward Action Requests

One Forward Action Request (FAR) was raised during this joint validation and verification.

3 VALIDATION FINDINGS

3.1 Project Details

The Project Activity “Solitude 16 MW solar PV” consists the installation of 59,376 solar polycrystalline modules with a total capacity of 16.328 MW in Solitude near the village of Triolet in the north of Mauritius, in the district of Pamplemousses. VVB has received soft copies of the Land lease document/10/, General layout of the plant/13/Yield Simulation Report/27/, ESPA/16/, Commercial Operation Date certificate/14/, Design Compliance Report/27/, Completion Certificate/15/, photographs of the actual site/22,23/ documents to verify the project implementation and operation.

Technical Details of the project activity

S.No.	Technical details of the equipment	Description	Means of Verification
1.	Technology Used	Solar PV technology (Poly-Crystaline)	Installation of 59,376 solar polycrystalline modules with a total capacity of 16.328 MW confirmed through CDM validation report/33/ and video call with PP representatives present on site which showed the validation team the plant site and the name plates of all the equipment and photographic evidence from the site/22,23/.
2.	Solar modules capacity and Model	Capacity:16.328 MW; Manufacturer: China Sunergy Co., Ltd Model: CSUN275-60P	The capacity, characteristics of solar modules and other technical description of the plant(inverters & transformers' shelter boxes) was confirmed through CDM validation report/33/ and video call with PP representatives present on site which showed the validation team the plant site and the name plates of all the equipment The technology specification/17/ shared by PP specifies the project technology modules' make, model, and capacity details and was found consistent with information provided in Joint PD and MR /1/.
3.	Inverter	Manufacturer: INGECON SUN Model:1220TL Number of Inverters: 11,	Specifications of the inverters installed at the project activity site were checked to confirm the numbers, make, and model of the inverter through photographs of name plates checked during video call with PP and manufacturer's specification/17/.
4.	Transformer	Manufacturer: SEA Model:A0Bk Number of Inverters: 4	As confirmed from technical specifications/17/ and registered CDM PDD/2/.

4.	Solar PV Life	25-year linear performance (limited warranty)	The specifications shared by PP state the lifetime of the equipment, which is 25 years as confirmed from technical specifications/17/ and registered CDM PDD/2/.

Metering System

S. No.	Meter Serial number	Specifications	Means of Verification
1.	16538506 (Main meter)	Make: EDMI Type:Mk6E Accuracy: 0.2s (±0.02%)	Bidirectional electricity meters at CEB substation. The meters measure the net electrical energy supplied from Voltas Yellow Ltd (deducing auxiliary consumption) to CEB. The meter calibration details/24/, manufacturer specifications/17/ and meter image/23/ shared by PP has provided VVB assessment team with sufficient evidence on project activity metering system. Calibration details are discussed in section 4.5 of this report.
2.	16538507 (Check meter)	Make: EDMI Type:Mk6E Accuracy: 0.2s (±0.02%)	Bidirectional electricity meters at CEB substation. The meters measure the net electrical energy supplied from Voltas Yellow Ltd (deducing auxiliary consumption) to CEB. The meter calibration details/24/, manufacturer specifications/17/ and meter image/23/ shared by PP has provided VVB assessment team with sufficient evidence on project activity metering system. Calibration details are discussed in section 4.5 of this report.

The power generated is sold the Central Electricity Board (CEB) grid thereby replacing its dependency on the generation of electricity fossil fuels significantly. Energy Supply Purchase

Agreement signed between CEB and Voltas Yellow Ltd. confirmed that the electricity is being sent to the Grid/16/.

The project aims to achieve 29,233 MWh annually resulting in the emission reductions of 28,984 tons CO2eq per year on average as confirmed from the ER sheet/5/.

As per the desk review, remote survey observations, and collected evidences, it was possible to assess that, in general, the project activity has been implemented as described in the Joint VCS PD and MR/1/.

There is no material discrepancy between the monitoring plan set out in the project description of Joint PD and MR and CDM PDD/1,2/ and the applied methodology/4/.

Some of the characteristics of the PA are mentioned below:

- Project Proponent: Voltas Yellow Ltd as confirmed from the ESPA/16/ and CDM validation report/33/.

- Other entities involved: AERA Group is the buyer of VERs and has a signed contract in place with the PP/36/.

- Project Category: Project since the emission reductions are less than 300,000 tonnes of Co_2e per year as checked from the ER sheet/5/.

- Estimated GHG reductions: 28,984 tCO_2 reductions on an average per year and 2,02,889 tCO_2 reductions during the entire crediting period of 7 years as checked from the ER sheet/5/. The calculations in the ER sheet were found to be accurate and appropriate.

- Start date: 19/12/2018, which is the which is the date of commissioning, synchronization and start of GHG emissions as confirmed from the Commercial Operation date issued by Republic of Mauritius/14/ and electricity generation data and power invoices/21/.

- Crediting period: The project crediting period starts on 19/12/2018 and lasts 7 years (fixed), until 18/12/2025.

- VERs Ownership: Voltas Yellow Ltd., as confirmed through the Power Purchase Agreement with CEB, dated 16/12/2016/16/.

- Project Location: The project is situated in in Solitude near the village of Triolet in the north of Mauritius, in the district of Pamplemousses. This was confirmed from the using Latlong coordinates application/28/.

- Conditions prior to project initiation: Electricity generation by fossil-fuel fired power plants as confirmed from CDM validation report/32/ and interview with the PP representative. The baseline situation is in-line with the applied methodology/4/,

- Compliance with applicable laws, statutes and other regulatory frameworks: Solitude 16 MW solar PV project has all necessary clearances in order to generate energy from Solar panels, in accordance with Mauritian legislation as confirmed through the EIA license/26/ and electricity generation license/14/ granted to the project.

- Emissions trading programs and other binding limits: GHG emission reductions generated by the project are not included in an emissions trading program or any other mechanism that includes GHG allowance trading as declared by Voltas Yellow Ltd. through self-declaration document/18/.

- Other forms of environmental credit sought or received and eligible to be sought or received: The project has not received nor sought any other form of environmental credit/18/.

- Participation under other GHG programs: The project activity is also registered under CDM (UNFCCC project number 10483) since 31/05/2019 but the project has not sought any carbon credits from CDM so far as confirmed from CDM project webpage/27/.

- Rejection by other GHG programs: the project was not rejected under any other GHG programs/18/.

- Additional Information: There is no commercially sensitive information or sustainable development contributions that are being claimed.

All information provided was verified from the remote survey and the supporting documents and evidences provided by the PP. Thus, VVB confirms that the description provided in project description is accurate, complete, and appropriately provides the understanding of the nature of the project. The project is found to be implemented in line with the project description in Joint PD and MR and CDM PDD/1,2/ and applied methodology/4/.

The PP has contracted the VVB for joint validation and verification of the Project Activity to VCS on 02/03/2020, therefore the gap validation of the Project Activity to VCS is considered valid and is eligible to be included under the scope of VCS Program (Version 3.0) as per requirement stated under Appendix II of VCS Standard Version 4/7/.

3.2 Participation under Other GHG Programs

The project activity is also registered under CDM (UNFCCC project number 10483) since 31/05/2019 /27/ but the project has not sought any carbon credits under CDM. It has been verified from the VCS registry/3/ and CDM webpage/27/ that no overlap concerning the dates for the current monitoring period has taken place. Furthermore, an undertaking has been procured from Voltas Yellow Ltd. to establish that “The project has been registered under another GHG program (CDM Project activity n°10483) but is not seeking any renewal or new registration under the same”/18/.

3.3 Safeguards

3.3.1 No Net Harm

No negative environmental and social impacts have been identified.

3.3.2 Local Stakeholder Consultation

The Project "Solitude 16 MW solar PV" has already been registered under CDM. The CDM PDD/2/ describes the local stakeholder consultation process in detail, by the Project Developer, in the registered CDM PDD/2/. A fresh LSC was not conducted by the CME again while requesting VCS gap validation as there has not been any change in the project description since the CDM registration of the project activity. The VVB corroborated the LSC conducted under CDM through CDM validation report/33/ and other evidences/20/ to ensure that the project has considered and included the feedback of all the relevant stakeholders during the implementation of the Project Activity. A local stakeholder meeting was conducted on 06/07/2018 in Solitude, District of Pamplemousses, Mauritius. Most of the stakeholders were either informed by invitation or by a press advertisement in the local newspaper/20(c)and 20(d)/.

The participation of relevant stakeholder was confirmed through stakeholder consultation attendance sheet/20(f)/ and photograph of group attending the meeting/20(b)/.

During the local stakeholder consultation meeting, all the comments were invited in an open and transparent manner following a presentation on the proposed project activity. The public comments received during the Local Stakeholder Consultation were answered during the meeting. No negative comment was received

From the table above, it can be confirmed that the Project Developers took due consideration of all the comments received during the Local Stakeholder Consultation Process. There is also an ongoing mechanism to record and address any comments received by the stakeholders during the course of implementation of the Project Activity. The CDM PDD/2/ states following way for stakeholders to communicate:

a) Direct communication at the solar farm's office

b) Electronic communication via the project owner's website or telephone number

c) Indirect communication through local and national government authorities and CEB.

However, no complaint has been registered through any of these means yet as confirmed through the interviews with the PP's representatives.

3.3.3 Environmental Impact

The Project Activity was required to undergo an Environmental Impact Assessment in accordance with Environmental Protection Act (EPA, 2008). The Project Developer carried out an elaborate Environmental Impact Assessment of the Project and an EIA License was granted by the department of Environment on 05/09/2017/26/.

The EIA Report/19/ discussed all the aspects like background of the Project, baseline environment, project description, assessment of impact and mitigating measures and the Environmental Monitoring Plan related to the Project Activity. The EIA report has summarized the findings during the Operational stage impacts in the following manner:

1. Toxic and hazardous materials used for production of PV cells
2. Atmospheric emissions of toxic substances due to incineration during decommissioning of PV systems
3. Occupational health hazards during manufacturing of PV cells
4. Public health hazard during manufacturing of PV cells
5. Soil and/or groundwater contamination due to improper disposal of batteries
6. Soil and/or groundwater contamination due to decommissioning of PV systems

The EIA Report/19/ has also devised various mitigation measures to minimize the impact of any adverse Environmental or Social hazards. The Validation team is therefore able to confirm that the Project is complying with all the Environmental norms in the host country and the issuance of the Environmental License has confirmed the same.

3.3.4 Public Comments

The Joint Project Description and Monitoring Report was made public from 08/09/2020 to 08/10/2020 during which no comments were received as confirmed from the project webpage/3/.

3.3.5 AFOLU-Specific Safeguards

Not Applicable.

3.4 Application of Methodology

3.4.1 Title and Reference

The Project Activity has applied the approved baseline and monitoring methodology ACM0002 version 20.0 - “Grid-connected electricity generation from renewable sources” which is the latest applicable version of the methodology/1,4/.

The methodology has referred the following tool for the calculation of various parameters:

“TOOL07: Tool to calculate the emission factor for an electricity system” version 7.0

3.4.2 Applicability

All applicability conditions of the valid version of the methodology (ACM0002 version 20) /4/ are met. All applicability conditions are completely and correctly included in the CDM PDD/2/ and referred in VCS joint PD and MR/1/.

Criteria	Means of Verification
This methodology is applicable to grid-connected renewable energy power generation project activities that: a) Install a Greenfield power plant; (b) Involve a capacity addition to (an) existing plant(s); (c) Involve a retrofit of (an) existing operating plants/units; (d) Involve a rehabilitation of (an) existing plant(s)/unit(s); or (e) Involve a replacement of (an) existing plant(s)/unit(s).	The project activity is greenfield solar power plant installed that supplied electricity to CEB as confirmed Energy Supply Purchase Agreement/16) land lease document/10/ and the Environment Impact Assessment Report /19/. The criterion is fulfilled.
The project activity may include renewable energy power plant/unit of one of the following types: hydro power plant/unit with or without reservoir, wind power plant/unit, geothermal power plant/unit, solar power plant/unit, wave power plant/unit or tidal power plant/unit;	The project activity is a solar photovoltaic as checked from video call with PP representatives present at the site, CDM validation report/33/ and EIA report/19/. Thus, the criterion is fulfilled.
In the case of capacity additions, retrofits, rehabilitations or replacements (except for wind, solar, wave or tidal power capacity addition projects the existing plant/unit started commercial operation prior to the start of a minimum historical reference period of five years, used for the calculation of baseline emissions and defined in the baseline emission section, and no capacity expansion, retrofit, or rehabilitation of the plant/unit has been undertaken between the start of this minimum historical reference period and the implementation of the project activity.	The project activity is a greenfield power plant as checked form the land lease agreement/10/, criterion is not applicable.
In case of hydro power plants, one of the following conditions shall apply: (a) The project activity is implemented in existing single or multiple reservoirs, with no change in the volume of any of the reservoirs; or (b) The project activity is implemented in existing single or multiple reservoirs, where the volume of the reservoir(s) is increased and the power density calculated using	The PA is not a hydro power plant. The criterion is not applicable.

equation (3) of the methodology ACM0002, is greater than 4 W/m^2; or (c) The project activity results in new single or multiple reservoirs and the power density, calculated using equation (3) of the methodology ACM0002, is greater than 4 W/m^2; or (d) The project activity is an integrated hydro power project involving multiple reservoirs, where the power density for any of the reservoirs, calculated using equation (3) of the methodology ACM0002, is lower than or equal to 4 W/m^2, all of the following conditions shall apply: - The power density calculated using the total installed capacity of the integrated project, as per equation (4) of the methodology ACM0002, is greater than 4 W/m^2; - Water flow between reservoirs is not used by any other hydropower unit which is not a part of the project activity; - Installed capacity of the power plant(s) with power density lower than or equal to 4 W/m^2 shall be: a.) Lower than or equal to 15 MW; and b.) Less than 10 per cent of the total installed capacity of integrated hydro power project.	
In the case of integrated hydro power projects, project proponent shall: - Demonstrate that water flow from upstream power plants/units spill directly to the downstream reservoir and that collectively constitute to the generation capacity of the integrated hydro power project; or - Provide an analysis of the water balance covering the water fed to power units, with all possible combinations of reservoirs and without the construction of reservoirs. The purpose of water balance is to demonstrate the requirement of specific combination of reservoirs constructed under CDM project activity for the optimization of power output. This demonstration has to be carried out in the specific scenario of water availability in different seasons to optimize the water flow at the inlet of power units. Therefore, this water balance will take into account seasonal flows from river, tributaries (if any), and rainfall for minimum five years prior to implementation of CDM project activity.	The PA is not a hydro power plant. The criterion is not applicable.

The methodology is not applicable to: - Project activities that involve switching from fossil fuels to renewable energy sources at the site of the project activity, since in this case the baseline may be the continued use of fossil fuels at the site; - Biomass fired power plants/units.	The project does not fall in either of the categories mentioned in this criterion. The criterion not applicable.
In the case of retrofits, rehabilitations, replacements, or capacity additions, this methodology is only applicable if the most plausible baseline scenario, as a result of the identification of baseline scenario, is "the continuation of the current situation, that is to use the power generation equipment that was already in use prior to the implementation of the project activity and undertaking business as usual maintenance".	The project activity is a greenfield power plant as checked form the land lease agreement/10/, criterion is not applicable.
In addition, the applicability conditions included in the tools referred to above apply.	There is only Tool 7 applied. The applicability conditions of included tool is assessed below.

Compliance with applicability conditions of "Tool to calculate emission factor for an electricity system":

Criteria	DOE Assessment
This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g. demand-side energy efficiency projects).	The project activity is greenfield solar power plant installed that supplied electricity to CEB as confirmed Energy Supply Purchase Agreement/16) land lease document/10/ and the Environment Impact Assessment Report /19/. The criterion is fulfilled.

Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off -grid power plants. In the latter case, two sub-options under the step 2 of the tool are available to the project participants, i.e. option IIa and option IIb. If option IIa is chosen, the conditions specified in "Appendix 2: Procedures related to off-grid power generation" should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	The PA involves the calculation of emission factor for grid power plants only. Thus, the criterion is met.
In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.	Mauritius is not an Annex I country as checked from list published online/37/. The tool is applicable for the PA under validation.
Under this tool, the value applied to the CO_2 emission factor of biofuels is zero.	The PA only involves no biofuel usage. The condition is not applicable.

3.4.3 Project Boundary

The project boundary is given by the applied methodology ACM0002, version 20.0/4/

"The spatial extent of the project boundary includes the project power plant and all power plants connected physically to the electricity system that the VCS project power plant is connected to."

The project activity consists of the installation of 59,376 solar polycrystalline photovoltaic modules. The total capacity of the project is 16.328 MW in which the power is off-taken and distributed to Central Electricity Board (CEB) grid. The project boundary includes the project activity site where the electricity is being produced and the grid that the power plant is connected to, which has been illustrated in the Section 1.12 of the Joint PD and MR Report/1,2/ and gives clear understanding of the project boundary; thus it is acceptable.

The consideration, by the PP, of only CO2 gas for the baseline emissions is conservative and in line with the methodology. The exclusion of CH4 & N2O in the baseline scenario is appropriate. The project activity involves the generation of electricity using solar energy. Hence, there are no project emissions associated with this project activity. Therefore, the exclusion of CO2, CH4 & N2O in the project scenario are appropriate. There are no other sources of project emissions. Hence, the project participant has considered the project emissions as zero for project activity; this is in line with the methodology.

The validation team is able to conclude that the project boundary and selected sources are applied as per the methodology /4/ and the applicable VCS criteria/7/.

3.4.4 Baseline Scenario

The project activity involves the installation of a newly built and grid-connected renewable power plant and supplying the generated electricity to Central Electricity Board (CEB) grid, hence, according to the applied ACM0002, version 20.0/4/, the baseline scenario is determined properly as:

"Electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in the "Tool to calculate the emission factor for an electricity system."/12/.

Assumption and data: An article published on 19/08/2019 states that "CEB currently produces 40 percent of the country's total power requirement from four thermal power stations and eight hydroelectric plants. The remaining 60 percent is purchased from independent power producers, mainly private generators from the sugarcane industry using bagasse and imported coal as"/38/. The Joint PD &MR under section 1.1 states, "Electricity in Mauritius is mainly generated from coal and heavy fuel oil"/1/. The information was checked from the sourced document/39/ and found to be correct.

The identified baseline for the project activity is the most likely scenario of what would have occurred in the absence of the project activity. The project participant has included all sources and references used for baseline determination for the project activity in the Joint PD and MR Report /1/ and the identified baseline is justified appropriately by the project participant.

Moreover, websites of Ministry Of Energy and Public Utilities And Ministry of Environment, Solid Waste Management and Climate Change were checked to confirm that No national or regional policy mandates or restricts the project activity.

The validation team confirms that:

(a) All the assumptions and data used by the PP are listed in the VCS Joint PD and MR Report/1/, including their references and sources.

(b) All documentation used is relevant for establishing the baseline scenario and correctly quoted and interpreted in the Joint PD and MR/1/.

(c) Assumptions and data used in the identification of the baseline scenario are justified appropriately, supported by evidence and can be deemed reasonable.

(d) Relevant national and/or sectoral policies and circumstances are considered and listed in the Joint PD and MR/1/.

(e)The approved baseline methodology has been correctly applied to identify the most reasonable baseline scenario and the identified baseline scenario reasonably represents what would occur in the absence of the proposed project activity.

3.4.5 Additionality

As per para 3.19.5(1) of the Verra Standard version 4.1, the additionality for the project has been considered same as was assessed at the time of CDM validation/33/ as this section need not to be completed during the gap-validation. The additionality of the project under CDM was conducted in-line to methodology ACM0002 Version 19.0 which was valid at that time.

According to paragraph 28 of the applied methodology ACM0002: Large-scale Consolidated Methodology: Grid-connected electricity generation from renewable sources Version 19.0/42/, a simplified procedure to demonstrate additionality is applicable to grid connected electricity generation Solar photovoltaic technology-based plant. A specific technology in the positive list is deemed automatically additional if the following conditions are met at the time of submission of CDM PDD/2/:

a) The percentage share of total installed capacity of the specific technology in the total installed grid connected power generation capacity in the host country is equal to or less than two per cent; or
b) The total installed capacity of the technology in the host country is less than or equal to 50 MW.

The Project Activity meets condition b) of the methodology.

The total installed capacity of solar PV in the host country at the time of CDM registration was 21.2 MW. This has been confirmed by the Validation team from The Ministry of Energy and Public Utilities' operational power plants list available on ministry's website/41/.

Since, the total installed solar PV capacity in Mauritius was far below the 50 MW threshold, the Project is automatically additional.

The positive list of technologies indicated in paragraph 28 above is valid till 30/08/2020 as per the applied methodology version 19 para 31/42/ and was found to be appropriate for this validation assessment because of the following condition:

- Verra Standard 4.1 para 3.19.5(1) states that for gap validation (project registered under CDM) additionality section need not to be completed and thus, considering the additionality same as the one considered during CDM registration was found to be acceptable. This project got registered under CDM before 30/08/2020 when the additionality conditions under ACM0002 Version 19.0 was valid.

3.4.6 Quantification of GHG Emission Reductions and Removals

The proposed activity has applied baseline methodology ACM0002, version 20.0

The baseline emissions have been calculated as follows:

$$BE_y = EG_{PJ,y} \times EF_{grid,CM,y}$$

Where:

BE_y	Baseline emissions in year y (t CO_2/yr)
$EG_{PJ,y}$	Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the project activity in year y (MWh/yr)
$EF_{grid,CM,y}$	Combined margin CO_2 emission factor for grid connected power generation in year y calculated using the latest version of the "Tool to calculate the emission factor for an electricity system" (t CO_2/MWh)

Calculation of $EG_{PJ,y}$:

$$EG_{PJ,y} = EG_{facility,y}$$

$EG_{PJ,y}$	Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the project activity in year y (MWh/yr)
$EG_{facility,y}$	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh/yr)

Calculation of $EF_{grid,CM,y}$:

This parameter is calculated using the tool, "Tool to calculate the emission factor for an electricity-system" (Version 07.0). Emission factor once calculated, shall remain fixed during the entire duration of the monitoring period. The tool provides procedure to determine the following parameters:

$EF_{grid,CM,y}$	Combined margin CO2 emission factor for the project electricity system in year y
$EF_{grid,BM,y}$	Build margin CO2 emission factor for the project electricity system in year y

$EF_{grid,OM,y}$	Operating margin CO2 emission factor for the project electricity system in year y

The tool has six steps for the calculation of combined margin (CM) emission factor:

Step 1: Identify the relevant electricity systems:

The relevant electric power system is the national grid of Mauritius which is managed by Central Electricity Board (CEB) and power plants are physically connected through transmission and distribution lines to the project activity.

Step 2: Choose whether to include off-grid power plants in the project electricity system (optional):

In line with para 28 and 29 of applied tool/12/, project participant has chosen to include only grid-connected power plants in the calculations.

STEP 3: Select a method to determine the operating margin (OM) method

PP has selected option (a) in accordance with para 38 of applied tool/12/ i.e. Simple OM, Based on the data requirement and important conditions set to be met in order to apply specific OM method, the choice of option (a) was found appropriate because low cost/ must run plants in the host country constitute less than 50% of the total generation of the system/6/. As demonstrated in PDD, a total share of all the low-cost/must run resources with a total share not exceeding 20.4% between 2015 and 2017.

The simple OM can be calculated from the two data vintages.

- Ex-ante option- the emission factor is calculated at the time of validation and doesn't need any monitoring and recalculate during the whole crediting period.
- Ex-post option- The emission factor is calculated annually during the monitoring period.

PP has selected the Ex-ante option for the calculation of the OM emission factor.

Therefore, the selection of option (a) was found to be the most appropriate.

STEP 4: Calculate the operating margin emission factor according to the selected method

According to the applied tool, simple OM emission factor ($EF_{OM,simple,y}$) can be calculated in two ways:

(a) Option A: Based on the net electricity generation and a CO_2 emission factor of each power unit; or

(b) Option B: Based on the total net electricity generation of all power plants serving the system and the fuel types and total fuel consumption of the project electricity system.

According to paragraph 47 (b) of the applied tool, option B can only be used if the necessary data for option A is unavailable, Only nuclear and renewable power generation are considered as low-cost/must-run power sources and the quantity of electricity supplied to the grid by these sources is known and if Off-grid power plants are not included in the calculation (i.e. if Option I has been chosen in Step 2). Since all of the conditions have been met, PP chose Option B to calculate simple OM emission factor.

In line with para 54 of tool07/12/, the formula used in revised PDD/1,2/ for calculation of operating margin emission factor was found to be appropriate, as given below:

$$EF_{grid,OMsimple,y} = \frac{\sum_i FC_{i,y} \times NCV_{i,y} \times EF_{CO2,i,y}}{EG_y}$$

$EF_{grid,OMsimple,y}$	Simple operating margin CO2 emission factor in year y (t CO2/MWh);
$FC_{i,y}$	Amount of fuel type i consumed in the project electricity system in year y (mass or volume unit);
$NCV_{i,y}$	Net calorific value (energy content) of fuel type i in year y (GJ/mass or volume unit);
$EF_{co2,i,y}$	CO2 emission factor of fuel type i in year y (t CO2/GJ);
EG_y	Net electricity generated and delivered to the grid by all power sources serving the system, not including low-cost/ must-run power plants/units, in year y (MWh);
i	All fuel types combusted in power sources in the project electricity system in year y; and
y	The relevant year as per the data vintage chosen in Step 3

The data related to each fossil fuel used to power the different power plants has been procured by CEB as confirmed from the Emission Factor calculation sheet/6/ and other local values of NCV and IPCC default values have been used to calculate Simple OM Emission Factor. Based on the obtained data and the final calculations reviewed from Emission Factor calculation sheet, the calculated OM is 1.0282 tCO2 MWh.

STEP 5: Calculate the build margin (BM) emission factor

To calculate build margin emission factor, tool07/12/ provisions two options for selection of vintage of data used. Since either of the choices do not require any pre-condition to be met before being selected, option 1 has been chosen by the PP.

Project participant has applied option 1 from para 72 of Tool 07, which says that build margin emission factor shall be updated based on the most recent information available on units already built. The most recent consolidated data published available was confirmed to be of 2017/2/.

PP has applied the steps defined in procedure for selection of power units in operation at the time of submission (defined in the Para 75 of Tool 07/12/), which has been checked and found to be valid for the calculation of BM emission factor.

The 5 power units considered for BM calculation, which have been appropriately selected as per the guidance in the para 75 of tool 7/12/, generate 20% of the system's energy, and the calculated build margin is the generation-weighted average emission factor (tCO_2/MWh) of all power units m during the most recent year y for which power generation data is available.
The methodological choices and calculations were found to be appropriate and acceptable by the assessment team.
The electricity generation data used for calculation of build margin emission factor was found to be the latest data available at the time of submission of PDD to the validating DOE.

The build margin emissions factor for which electricity generation data is available (2017 in present case), calculated as follows:

$$EF_{grid,BM,y} = \frac{\sum_m EG_{m,y} \times EF_{EL,m,y}}{\sum_m EG_{m,y}}$$

$EF_{grid,BM,y}$	Build margin CO_2 emission factor in year y (t CO_2/MWh)
$EG_{m,y}$	Net quantity of electricity generated and delivered to the grid by power unit m in year y (MWh)
$EF_{EL,m,y}$	CO_2 emission factor of power unit m in year y (t CO_2/MWh)
m	Power units included in the build margin
y	Most recent historical year for which electricity generation data is available

Based on the obtained data and the final calculations reviewed from Emission Factor calculation sheet/6/, the calculated BM is 0.8814 tCO_2 MWh.

Step 6: Calculate the combined margin emission factor

The combined margin emissions factor is calculated as follows:

$EFgrid, CM,y = EFgrid,OM,y \times wOM + EFgrid,BM,y \times wB$M

$EF_{grid,BM,y}$	Build margin CO_2 emission factor in year y (t CO_2/MWh)
$EF_{grid,OM,y}$	Operating margin CO_2 emission factor in year y (tCO_2/MWh)
wOM	Weighting of operating margin emissions factor (%); and
wBM	Weighting of build margin emissions factor (%).

The default values of wOM(0.75) and wBM(0.25) have been used and based on the Based on the obtained data and the final calculations reviewed from Emission Factor calculation sheet/6/, the calculated CM is 0.9915 tCO_2 MWh.

3.4.7 Methodology Deviations

No deviation from methodology and tools was requested during the validation process.

3.4.8 Monitoring Plan

According to the applied methodology: ACM0002/4/, the only parameter to be monitored is the “Quantity of net electricity generation supplied by the project plant/unit to the grid in year y”

This parameter will be calculated as difference between

(a) The quantity of electricity supplied by the project plant/unit to the grid; and

(b) The quantity of electricity received by the project plant/unit from the grid.

These will be monitored using bi-directional energy meter (main and check meters) calibrated and cross-checked with power invoices as per monitoring reconciliation practices with CEB.

The monitoring methodology/4/ applies consistently the choice of the option selected for monitoring of baseline emissions. The monitoring plan provide procedures for the collection and archiving of all relevant data necessary for estimation or measuring the emission reductions within the project boundary during the crediting period.

Meter Laboratory of CEB will be solely responsible for the selection, installation, calibration, servicing, testing and repairing of all CEB energy meters. The Joint PD and VCS MR/1/ has been reviewed to check that the procedure for data uncertainty, emergency preparedness, roles and responsibility, operational and management structure. The details of the metering units are already mentioned and corroborated under section 3.1 of this report. The monitoring plan completely describes all measures to be implemented for monitoring all parameters required. The validation team confirms that:

1. The monitoring plan included in the CDM PDD/2/ and Joint PD and MR/1/ is based on the baseline methodology ACM0002 version 20.0/4/ which has been applied to the proposed VCS project activity

2. The monitoring arrangements described in the monitoring plan are feasible within the project design.

3.5 Non-Permanence Risk Analysis

No non-permanence risk has been identified.

4 VERIFICATION FINDINGS

4.1 Accuracy of GHG Emission Reduction and Removal Calculations

The project monitoring has been carried in accordance with the registered CDM PDD/2/ and VCS Joint PD and MR/1/. The assessment team has verified the information flow (from data generation, aggregation, to recording, calculation and reporting for these parameters including the values).

The emission reductions are based on the fuel savings by the project activity.

The emission reduction calculation is accurate from the point of data collection, transfer and the formulas applied are in accordance with the applied methodology/4/ and CDM registered PDD/2/.

The emission reduction calculations have been followed in accordance with the registered CDM PDD/2/ and applied methodology/4/.

The default values applied for calculations were consistent to the registered CDM PDD/2/ and Joint PD and MR/1/.

The data transfer from data generation sheets to final emission reduction calculation sheet/5/ has been transparently described and followed. The final values are reproducible, and all the calculation are linked and clearly presented in the emission reduction calculation sheet/5/.

The assessment team is in a position to conclude that the monitoring for the concerned period has been done in accordance with the procedures laid in the registered documents and the resulting emission reductions are measurable and conservative.

4.1.1. Parameter(s) available at the time of Validation

The table given below describe how the ex-ante parameter $EF_{grid,CM,y}$, which is to be measured according to the monitoring plan, has been verified to confirm that the actual monitoring complies with the monitoring plan, monitoring data has been thoroughly assessed and that the sampling requirement are met.

Ex-Ante Parameter	Assessment
$EF_{grid,CM,y}$ Combined margin CO_2 emission factor for grid connected power generation in year y calculated using the latest version of the “Tool to calculate the emission factor for an electricity system”	PP has applied a value of 0.9915 tCO_2/MWh as per Tool 7: “Tool to calculate the emission factor for an electricity system”/12/.

	The calculation related to the parameter has been described in detail under Section 3.4.6 of the report. The Joint PD and MR/1/ refers to CDM PDD/2/. The value is consistently applied in the EF sheet/6/.
EFgrid,OM,y Operating Margin CO2 emission factor for grid connected power generation in year y calculated using the latest version of the "Tool to calculate the emission factor for an electricity system"	PP has applied a value of 1.0282 tCO_2/MWh as per Tool 7: "Tool to calculate the emission factor for an electricity system"/12/. The calculation related to the parameter has been described in detail under Section 3.4.6 of the report. The Joint PD and MR/1/ refers to CDM PDD/2/. The value is consistently applied in the EF sheet/6/.
EFgrid,BM,y Build Margin CO2 emission factor for grid connected power generation in year y calculated using the latest version of the "Tool to calculate the emission factor for an electricity system"	PP has applied a value of 0.8814 tCO_2/MWh as per Tool 7: "Tool to calculate the emission factor for an electricity system"/12/. The calculation related to the parameter has been described in detail under Section 3.4.6 of the report. The Joint PD and MR/1/ refers to CDM PDD/2/. The value is consistently applied in the EF sheet/6/.
The percentage share of total installed capacity of the solar PV in the total installed grid connected power generation capacity in the host country.	PP has applied a value of 2.6% as per Statistics Mauritius (2018): Digest of Water and Energy Statistics -2017, Port Louis as checked from the government published data/29/.
The total installed capacity of solar PV	PP has applied a value of 21.2MW The value has been sourced from the website of Mauritian Government/41/.

4.1.2. Monitored Parameter(s)

<table>
<tr><td>Parameter</td><td>$EG_{facility,y}$
Quantity of net electricity generation supplied by the project plant/unit to the grid in year y</td></tr>
<tr><td>Means of verification</td><td>
<table>
<tr><th>Criteria/Requirements</th><th>Assessment/Observation</th></tr>
<tr><td>Measuring /Reading /Recording frequency</td><td>The parameter is monitored continuously for the estimation of Voluntary Emission Reductions. The PP is required to monitor the parameter at least monthly. The parameter here is being monitored continuously through the Electricity meter(s) present at the project site.</td></tr>
<tr><td>Is measuring and reporting frequency in accordance with the monitoring plan and monitoring methodology? (Yes / No)</td><td>Yes, measuring and reporting frequency is in accordance with monitoring plan /1,2/ and methodology/4/.</td></tr>
<tr><td>Monitoring equipment</td><td>Electricity outputs are recorded by bidirectional electricity meters at the project site of make EDMI and type MK6E/23,17/. The precision of the meters is 0.2s as checked from the meter specifications provided by the PP/17/.</td></tr>
<tr><td>Is accuracy of the monitoring equipment as stated in the monitoring plan? If the monitoring plan does not specify the accuracy of the monitoring equipment, does the accuracy of the monitoring equipment comply with local/national standards, or as per the manufacturer's specification?</td><td>As per the monitoring plan/1,2/, the accuracy of the electricity meters is 0.2S.

The same has been confirmed from the meter's manufacturer's specification/17/ and through the calibration certificate document provided by CEB/24/.</td></tr>
</table>
</td></tr>
</table>

	Is the accuracy valid for the entire measuring range or do different accuracy levels apply to different measuring ranges?	As per the monitoring plan/1,2/ and the manufacturer's specifications/17/, the accuracy of 0.2S is valid for the entire measuring range.
	Calibration frequency /interval:	The electricity meters are already calibrated at the time of manufacturing and as per the manufacturer's specifications/17/, the meters are not required to be calibrated thereafter. However, as confirmed from the ESPA/16/, CEB shall inspect each CEB meter upon installation and at least once every year thereafter. CEB is also required to test the certification of the meters through an accuracy test at least once every four years. CEB performed an initial verification of the electricity meters on 01/10/2018 and concluded that the accuracies of the meters are within the standard. This has been confirmed from the meter test certificate issued by CEB to Voltas Yellow Ltd. /24/.
	Is the calibration interval in line with the monitoring plan and/or methodology? If the monitoring plan does not specify the frequency of calibration, is the selected frequency in accordance with the local/national standards, pending until the findings are closed or as per the manufacturer's specifications?	As per the monitoring plan, the electricty meters are already calibrated at the time of installation and the same has been checked from the manufacturer specifications provided by the PP/17/. However, as established in the ESPA/16/, CEB performed an initial verification of the electricity meters on 01/10/2018 and concluded that the accuracies of the meters are within the standard. This has been confirmed from the meter test certificate issued by CEB to Voltas Yellow Ltd. /24/.
	Is(are) calibration(s) valid for the whole reporting period?	Although no calibration of the meters is required to be done, a verification of the meters was conducted on 01/10/2018/24/. This has been confirmed from meter test

		certificate issued by CEB to Voltas Yellow Ltd./24/.
	How were the values in the monitoring report verified?	The final value calculated for $EG_{facility,y}$ are as follows: <table><tr><th>Year</th><th>2018</th><th>2019</th><th>2020</th><th>Total</th></tr><tr><td>Value</td><td>885 MWh</td><td>21,251 MWh</td><td>7,800 MWh</td><td>29,936MWh</td></tr></table> The following monitored values were checked against the power invoices/21/ and the ER sheet/5/ provided by the PP to the verification team.
	Does the data management ensure correct transfer of data and reporting of emission reductions and are necessary QA/QC processes in place?	Yes, QA/QC procedures were found to be in place.
	In case project participants have temporarily not monitored the parameter, has either i) a deviation been approved by the CDM EB or ii) has the parameter been estimated as stipulated by Appendix 1 to the CDM Project Standard?	No such issue
Findings	No findings were raised.	
Conclusion	The parameter has been monitored appropriately, in accordance with the registered monitoring plan (as per measurement methods and procedures to be applied) and applied methodology. The monitoring results were recorded consistently as per the approved frequency in the monitoring plan.	

	The verification team is able to confirm that the project is implemented as per the project description(in Joint PD and MR and CDM PDD) and there is no discrepancy observed between the actual monitoring system and the monitoring plan. Monthly values of electricity generated inserted in the ER sheet/5/ was verified with the import and export invoices provided by the project proponent/21/. Since 100% data was verified, the team can ascertain that the values taken for emission reduction calculation are free from material errors.

4.2 Quality of Evidence to Determine GHG Emission Reductions and Removals

The emission reductions calculations for this verification were verified against many evidences: Electricity generation invoices raised to CEB- signed by both PP and CEB representatives/21/, meter verification certificates by CEB/24/. The evidences checked were found to be appropriate and reliable sources of information. The evidences used were approved by third parties and therefore, found to be non-biased and appropriate. The information was also cross checked through received documents and interviews conducted and observations made during the remote survey conducted by the verification team.

The data was found to be appropriately stored (achieved for at least two years) and easily retrievable if required.

Verification of meters: Although calibration of meters is not required for this project as per meter requirements, an annual verification frequency is followed by CEB to check the functioning of the meters. This was found to be implemented in accordance with Joint PD and MR and CDM PDD/1,2/.

The baseline emissions were calculated in line to the applied methodology ACM0002: Large-scale consolidated Methodology: Grid-connected electricity generation from renewable sources, Version 20/4/.The verification team confirms that appropriate methods and formulae for calculating baseline emissions have been followed. The assumptions, emission factors and default values that were applied in the calculations are justified.

All the data were made available and have been monitored as per required monitoring frequency. The means of verification for the values of parameters, used for baseline emission calculation, is described above in previous sections.

5 VALIDATION AND VERIFICATION CONCLUSION

Earthood Services Private Limited (Earthood), contracted by Voltas Yellow Ltd., has performed the joint verification and validation of the emission reductions for the VCS project activity Ref number 2308 “Solitude 16 MW solar PV” in Mauritius for the crediting period **19/12/2018 to 18/12/2025** and monitoring period **19/12/2018 to 30/04/2020** as reported in the Joint VCS PD and MR version 1.2 dated 08/10/2020. The company, Voltas Yellow Ltd. is responsible for the collection of data in accordance with the monitoring plan and the reporting of GHG emissions reductions from the project activity. It is our responsibility to express an independent validation and verification statement on the reported GHG emission reductions from the project activity.

Earthood commenced the validation and verification based on the baseline and monitoring methodology ACM0002 version 20.0, the monitoring plan contained in the Joint Project Description and Monitoring Report Version 1.2 dated 08/10/2020 and VCS guidelines version 4.0, as per the process described under Section 2 of this report.

Earthood’s validation and verification approach is based on the understanding of the risks associated with reporting of GHG emission data and the controls in place to mitigate these. Earthood planned and performed the validation and verification by obtaining evidence and other information and explanations that Earthood considered necessary to give reasonable assurance that reported GHG emission reductions are fairly stated.

In our opinion the GHG emissions reductions reported for the project activity for the period 19/12/2018 to 30/04/2020 are fairly stated in the Joint Project Description and Monitoring Report Version 1.2 dated 08/10/2020. The GHG emission reductions were calculated correctly based on the approved baseline and monitoring methodology ACM0002 version 20.0, and the VCS standard.

Verification period: From 19-December-2018 to 30-April-2020

Verified GHG emission reductions and removals in the above verification period:

Year	Baseline emissions or removals (tCO2e)	Project emissions or removals (tCO2e)	Leakage emissions (tCO2e)	Net GHG emission reductions or removals (tCO2e)
2018	878	-	-	878
2019	21,070	-	-	21,070
2020	7,733	-	-	7,733
Total	29,681	-	-	29,681

Approved by

Dr. Kaviraj Singh

Managing Director

Earthood Services Privated Limited

Date: 20/10/2020

Place: Gurgaon, Haryana

APPENDIX I: <REFERENCES>

No.	Title	References
1.	VCS-Joint Project Description and Monitoring Report	Version 1.2 Dated: 08/10/2020
2.	CDM Registered PDD	Version 1.4 Dated: 02/05/2019
3.	https://registry.verra.org/app/projectDetail/VCS/2308	-
4.	Methodology: ACM0002: Large-scale Consolidated Methodology: Grid-connected electricity generation from renewable sources	Version 20.0
5.	Ex-ante and Ex-post ER Spreadsheet – final	Corresponding to final version of Joint PD and MR
6.	Grid Emission factor calculation sheet	Corresponding to final version of CDM PDD
7.	VCS Standard	Version 4.0
8.	VCS Validation and Verification Manual https://verra.org/wp-content/uploads/2018/03/VCS_Validation_Verification_Manual_v3.2.pdf	Version 3.2
9.	O&M reports	-
10.	Land lease agreement	-
11.	GreenYellow Project producible note	-
12.	Tool 07: Tool to calculate the emission factor for an electricity system	Version 07.0

13.	General Plant Layout	-
14.	Commercial Operation Certificate	Dated: 18/12/2019
15.	Completion Certificate	30/11/2018
16.	CEB Voltas Yellow Ltd. Energy Supply Purchase Agreement	16/12/2016
17.	Manufacturer's and technical specification of the installed technology	Various
18.	Undertaking for non-participation in any other Environmental Credit Program except for CDM	15/06/2020
19.	ESIA Report	February'17
20.	LSC Report a. minutes of meeting b. Annex 1: Photo of the Group at the Stakeholder Consultation Meeting c. Annex 2: Invitation card to stakeholders meeting d. Annex 3: Notice for Local Stakeholder Consultation Meeting for CDM project in Lexpress newspaper Wednesday 27 June 2018 e. Annex 4: Power point presentation for Stakeholder's Consultation Meeting f. Annex 5: SC Attendance sheet	06/07/2018
21.	Monthly power import/export invoices	Various
22.	Photographic evidences from the project site	-
23.	Photographic evidence for the installed technology	-
24.	Meter calibration certificate (CEB)	01/10/2018
25.	EPC Invoice	Various
26.	EIA License	05/09/2017
27	PA 10483 https://cdm.unfccc.int/Projects/DB/RWTUV1557984446.68/view	-
28	https://www.latlong.net/c/?lat=-20.348404&long=57.552151	-
29	Statistics Mauritius (2018): Digest of Water and Energy Statistics -2017, Port Louis http://publicutilities.govmu.org/	-
30	https://verra.org/covid-19-travel-guidance/	18/03/2020

31	Remote survey video recording	18/09/2020
32	International Travel Ban- India https://boi.gov.in/content/advisory-travel-and-visa-restrictions-related-covid-19-1#:~:text=1.,flights%20especially%20approved%20by%20DGCA. https://timesofindia.indiatimes.com/videos/news/covid-19-international-flights-to-remain-suspended-till-october-31/videoshow/78419463.cms	24/05/2020 01/10/2020
33	CDM Validation report	Version 1.2 13/05/2019
34	https://www.nytimes.com/interactive/2020/world/asia/india-coronavirus-cases.html	Last accessed on 20/10/2020
35	Design Compliance Report	13/12/2018
36	VOLUNTARY EMISSION REDUCTIONS (VERs) – PURCHASE INDICATIVE TERM-SHEET signed between VERRA and Voltas Yellow	-
37	List of Annex 1 countries: https://www.oecd.org/env/cc/listofannexicountries.htm	Last accessed on 20/10/2020
38	Mauritius Energy – commercial guide: https://legacy.export.gov/article?id=Mauritius-Renewable-Energy#:~:text=CEB%20currently%20produces%2040%20percent,using%20bagasse%20and%20imported%20coal.	Last accessed on 20/10/2020
39	Ministry of Finance and Economic Development – Statistic Mauritius: https://unstats.un.org/unsd/environment/Compendia/Mauritius,%20Digest%20of%20Environment%20Statistics,%202017.pdf	Last accessed on 20/10/2020
40	Republic of Mauritius (fovernment website): a) https://govmu.org/EN/infoservices/energyutilities/Pages/default.aspx b) https://govmu.org/EN/infoservices/environment/Pages/default.aspx	Last accessed on 20/10/2020
41	DIGEST OF ENERGY AND WATER STATISTICS - 2018 https://statsmauritius.govmu.org/Documents/Statistics/Digests/Energy_Water/Digest_Energy_Yr18.pdf https://publicutilities.govmu.org/Pages/Energy%20Sector/EnergySector.aspx	Last accessed on 20/10/2020
42	For additionality: ACM0002: Large-scale Consolidated Methodology: Grid-connected electricity generation from renewable sources	Version 19.0
43	Contractual agreement between DOE and PP	Dated: 02/03/2020

APPENDIX II: <ABBREVIATIONS>

Abbreviations	Full texts
BE	Baseline Emission
CA	Corrective Action / Clarification Action
CAR	Corrective Action Request
CDM	Clean Development Mechanism
CEB	Central Electricity Board
CO_2	Carbon dioxide
CO_2e	Carbon dioxide equivalent
CL	Clarification Request
VVB	Validation and Verification Body
ER	Emission Reduction
ESPL	Earthood Services Private Limited
ERPA	Emission Reductions Purchase Agreement
FAR	Forward Action Request
GHG	Greenhouse gas(es)
MP	Monitoring Plan
MR	Monitoring Report
PA	Project Activity
PE	Project Emission
PP	Project Participant
QA/QC	Quality Assurance / Quality Control
UNFCCC	United Nations Framework Convention on Climate Change
VCS	Verified Carbon Standard
VCS-PD	VCS – Project Description
VCU	Verified Carbon Unit

XLS	Emission Reduction Calculation Spread Sheet
SLD	Single Line Diagram
JMR	Joint Meter Reading
PPA	Power Purchase Agreement
ESIA	Environment and Social Impact Assessment
ESMP	Environment and Social Management Plan
ESPA	Energy Supply Purchase Agreement

APPENDIX III: COMPETENCE OF TEAM MEMBERS AND TECHNICAL REVIEWERS

Competence Statement			
Name	Deepika Mahala		
Country	India		
Education	M. Sc. (Environmental Management), GGSIP University B.Sc. Hons. (Chemistry), Sri Venkateshwar College, DU		
Experience	3 Years +		
Field	Climate Change		
Approved Roles			
Team Leader	YES		
Validator	YES		
Verifier	YES		
Methodology Expert	ACM0002, AMS.I.D., AMS.I.A, AMS.III.AV, AMS.II.G		
Local expert	YES (India)		
Financial Expert	NO		
Technical Reviewer	YES		
TA Expert	YES (TA 1.2 & TA 3.1)		
Reviewed by	Shreya Garg	**Date**	14/09/2018
Approved by	Anshika Gupta	**Date**	14/09/2018

Competence Statement

Name	Shreya Garg		
Country	India		
Education	M.Sc. (Climate Science & Policy), TERI University		
Experience	6 Years +		
Field	Climate Change		
Approved Roles			
Team Leader	YES		
Validator	YES		
Verifier	YES		
Methodology Expert	AMS.I.A., AMS.I.C., AMS.I.D., AMS.I.F., AMS.II.D., AMS.II.G., AMS.II.J., AMS.III.AV., ACM0002, ACM0012		
Local expert	YES (India)		
Financial Expert	NO		
Technical Reviewer	YES		
TA Expert	YES (TA 1.2, TA 3.1)		
Reviewed by	Abhishek Mahawar	**Date**	01/03/2018
Approved by	Ashok Gautam	**Date**	01/03/2018

Competence Statement			
Name	Rahi Sahni		
Education	M.Sc Environment Science and Technology, Bharati Vidyapeeth University, Pune		
Experience	6 months		
Field	Climate Change and Environment		
Approved Roles			
Team Leader	NO		
Validator	Yes		
Verifier	Yes		
Methodology Expert	NO		
Local expert	NO		
Financial Expert	NO		
Technical Reviewer	NO		
TA Expert	NO		
Reviewed by	Shreya Garg	**Date**	09/04/2020
Approved by	Anshika Gupta	**Date**	09/04/2020

APPENDIX IV: FINDINGS

Table 1. CL from this verification

CL ID	01	**Section no.**	3.1	**Date :** 02/09/2020

Description of CL	
1. The overall capacity of the plant as mentioned in the Joint PD and MR is 16.328 MW. However, the capacity of the plant as agreed in the ESPA provided (page 3) is 13.60MWac and 15 MW in EIA report. 2. The end date of the monitoring period as mentioned in the Joint PD and MR (version 1.1) is 31/05/2020. However, as per the contract signed with ESPL, the end date of the monitoring period is 30/04/2020	
Project participant response	**Date :** 08/09/2020
1. 16.328 MW is the DC installed capacity of the solar PV modules; yet as per ESPA Schedule A p.82 "Maximum Installed Capacity of the Solar Facility shall be 16.344 MW_{DC}", while the EIA report didn't specify the distinction. *2. Monitoring period cut-off has been reverted to 30/04/2020*	
Documentation provided by project participant	
Revised PD/MR	
DOE assessment	**Date:** 13/09/2020
1. The Contracted/authorized power was checked to confirm the justification provided by the PP and was found to support it. The ESPA (page 82) states that the maximum installed capacity of the solar facility shall be 16.344 MW_{DC}. The finding is now closed. 2. PP has now revised the end date of the monitoring period to 30/04/2020 which is in line to the contract signed with the VVB. CL#01 is closed.	

Table 2. CAR from this verification

CAR ID	01	**Section no.**	3.4	**Date :** 02/09/2020
Description of CAR				
The latest applicable version of the methodology: ACM0002: Large-scale Consolidated Methodology: Grid-connected electricity generation from renewable sources, is 20.0. The Version 19 as applied by the PP expired on 24/07/2020 23:59:59 GMT				
Project participant response				**Date :** 08/09/2020
ACM0002 version updated to 20.0				
Documentation provided by project participant				
Revised PD/MR & xls				
DOE assessment				**Date:** 13/09/2020
PP has now applied the latest applicable version of the methodology i.e., version 20.0. The finding is now closed.				

CAR ID	02	**Section No.**	4.2	**Date :** 03/09/2020
Description of CAR				
1. The values of "BE_y" and $EG_{facility,y}$" for March, April and May'2020 reported in the table under Section 6.2 of the Joint PD and MR (Version 1.1) are inconsistent with ER sheet "ex-post MR1 2. The final ER value reported in the table under Section 6.2 of the Joint PD and MR (Version 1.1) is inconsistent with ER sheet "ex-post MR1, cell H:24 3. The values reported under Section 6.5 of the Joint PD and MR (Version 1.1) are inconsistent with the with ER sheet "ex-post				
Project participant response				**Date :** 08/09/2020

Corrected xls & PD/MR submitted	
Documentation provided by project participant	
Corrected xls & PD/MR	
DOE assessment	**Date:** 13/09/2020
1. The values of "BE_y" and $EG_{facility,y}$" for March and April have now been revised and are now consistent with the ER sheet. 2. The final value of ER has now been made consistent with the ER sheet "ex-post MR1, cell H:22" 3. The values reported under Section 6.5 of the Joint PD and MR (Version 1.2) are now consistent with the ER sheet The finding is now closed	

Table 3. FAR from this verification

FAR ID	01	**Section No.**	2.5.1 and 2.4	**Date :** 12/10/2020
Description of FAR				
The VVB involved in the next verification shall confirm the project implementation of PA in line with project description through on-site assessment.				
Project participant response				**Date :** DD/MM/YYYY
XX				
Documentation provided by project participant				
XX				
DOE assessment				**Date:** DD/MM/YYYY
XX				